

Ansh Bhansali

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EDUCATION

University of Illinois Urbana-Champaign, IL, USA

Aug 2025 – Jul 2026

Master of Engineering in Autonomy and Robotics | GPA: 3.9/4.0

Coursework: Humanoids, Robot Planning, Control Systems, Deep Learning, Computer Vision, Safe Autonomy

Sardar Patel Institute of Technology, Mumbai, India

Dec 2021 – Jul 2025

Bachelor of Technology, Electronics and Telecommunication | GPA: 3.9/4.0

SKILLS

- **Languages:** Python, C++, MATLAB, Java, Bash
- **AI & Perception:** VLA Models, Vision Transformers, PyTorch, RL (PPO), Computer Vision (YOLO, SAM), 6D Pose Estimation
- **Planning & Control:** Motion Planning (A*, MPC), Trajectory Optimization (ProxQP, CHOMP), SLAM, Sensor Fusion, CBFs
- **Tools & Frameworks:** ROS2, MuJoCo, Gazebo, Isaac Sim, Pinocchio, MoveIt, Git, Linux

EXPERIENCE

Research Engineer - IHSI, University of Illinois, IL, USA

Apr 2026 – Present

Project: ARPA-H MarginDx, AI-Guided Robotic Microsurgery for Tumor Margin Assessment

- Developing real-time **impedance force control** stack for **KUKA LBR Med 7 + PI hexapod** achieving tissue contact regulation.
- Built multi-layer IPC architecture (**Java/Python/UDP**) bridging KUKA, PC backend, and hexapod with **<10ms** command latency.
- Achieved **20μm** resolution 3D tissue surface reconstruction with stable contact force during raster scans on tissue phantoms.

Embedded Engineering Intern - Dimension Six Technologies, Mumbai, IN

May 2024 – May 2025

Project: STM32-based E-Bike Conversion Kit

- Engineered production-ready firmware with battery management and regenerative braking, extending vehicle range by **40%**.
- Designed a 4-layer PCB in KiCad for a custom **ESC**, optimizing power stage layout to reduce losses by **15%** under peak load.
- Deployed IoT stack on **ESP32S3** with RFID authentication, enabling remote monitoring and contactless payments over MQTT.

Robotics Research Intern - Indian Institute of Technology, Bombay, Mumbai, IN

Jan 2024 – Jun 2024

Project: Autonomous Rugged Robot for Military Applications

- Built **SLAM**-based autonomous navigation stack with path planning, achieving **95%** accuracy in dynamic indoor environments.
- Fused IMU, GPS, and RGB-D camera data via **Extended Kalman Filter**, improving localization accuracy by **20%**.
- Deployed **YOLOv3** for real-time human detection and designed an adaptive gait controller for robust stair-climbing traversal.

PROJECTS

Embodied Vision-Language-Action (VLA) Kitchen Agent (*CLIP, Whisper, GPT-4V, AI2-THOR*) ([GitHub](#))

- Built end-to-end VLA converting unseen **YouTube** cooking videos to robot commands, achieving **77-85%** accuracy.
- Trained **19M**-parameter Transformer on 1.6k+ clips; generated novel Indian-cooking dataset.
- Attained **90%+** precision on key primitives, **60-70%** zero-shot transfer to Western cuisine, with **85-90%** success in AI2-THOR.

CrossViT Cross-Embodiment Visual Perception (*ViT, PyTorch, Contrastive Learning*) ([Github](#))

- Trained cross-attention **Vision Transformer** conditioned on 32-dim kinematic features for **embodiment-aware** representations.
- Achieved **99.3-100%** embodiment classification accuracy vs 20% random chance baseline through kinematic fusion.
- Demonstrated **zero-shot generalization** to unseen humanoid robots with morphologically meaningful embedding structure.

Humanoid Whole-Body Motion Planning (*RL, MPC, Unitree G1, MuJoCo*) ([Github](#))

- Implemented motion planning stack using **ZMP** preview control, **A*** footstep planning, and **MPC** balance optimization
- Achieved **49%** lower energy use than PD control with MPC balance control, while remaining stable under **70-80N** perturbations.
- Integrated **RL**-based locomotion with Jacobian IK manipulation for coordinated walk-and-reach tasks, achieving **75%+** success.

VLM-Based Autonomous Navigation System (*GEM e4, GPT-4o, ROS2*) ([Github](#))

- Built autonomous navigation stack integrating **GPT-4o VLM** for intent-based planning with LiDAR-camera sensor fusion.
- Implemented full pipeline: BEV perception, A* path planning, and Stanley control achieving end-to-end task success.
- Achieved **88%** accuracy on KITTI real-world dataset for VLM-based goal localization from natural language commands.

Open-Vocabulary 6D Pose Estimation (*FoundationPose, SAM-3, Moondream2*) ([Report](#))

- Developed **zero-shot** 6D tracking architecture with VLM-based **semantic inventory** and Gemini API query enrichment.
- Integrated **SAM-3** prompt segmentation with **Objaverse** mesh retrieval, achieving **76.5-100%** ADD-S AUC on YCBV dataset.
- Built RGB-D variant using **LangSAM** with **InstantMesh** and median-volume ensemble selection for mesh consistency.